Abhilash Kumar Giri

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LinkedIn | Github | Portfolio

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience developing cloud-native applications, event-driven architectures, and scalable backend systems. Hands-on experience building Generative AI and Large Language Model (LLM) applications, including Retrieval-Augmented Generation (RAG), semantic search, prompt engineering, and LLM API integration. Proficient in Python, Node.js, AWS, vector search, and backend development, with a strong foundation in cloud infrastructure, system reliability, and automation.

SKILLS

- **Languages:** Python, JavaScript, TypeScript, SQL, Java
- **Backend & APIs:** Node.js, Express.js, FastAPI, Fastify, REST APIs, GraphQL, Webhooks
- **Cloud & AWS:** Lambda, API Gateway, S3, EC2, ECS, EKS, ECR, DynamoDB, SQS, SNS, EventBridge
- **Databases & Caching:** PostgreSQL, MySQL, SQLite, Redis, Supabase
- **AI & LLM Engineering:** RAG, FAISS, BM25, Reciprocal Rank Fusion (RRF), LLM Integration, Prompt Engineering, Transformer Architecture
- **DevOps & Monitoring:** Docker, Kubernetes, Terraform, AWS SAM, GitHub Actions, GitLab CI/CD, Azure DevOps, Datadog, CloudWatch

PROFESSIONAL EXPERIENCE

AbsoluteLabs **Apr. 2025 – Present**
Senior Software Developer *Hyderabad, IND*

- Engineered event-driven backend systems using AWS Lambda, API Gateway, SQS, SNS, S3, and DynamoDB to support scalable enterprise integrations.
- Integrated Newstore with Talon.One and Business Central via MuleSoft and Azure Blob Storage.
- Developed order-processing workflows using GraphQL APIs, Lambda, and Ometria Email API.
- Designed FIFO-based SQS pipelines for reliable, ordered message processing.
- Collaborated with cross-functional teams on system design, production troubleshooting, CI/CD workflows, and cloud infrastructure reliability.

AbsoluteLabs **Oct. 2023 – March 2025**
Software Developer *Hyderabad, IND*

- Developed and maintained serverless applications using AWS Lambda and S3.
- Optimized complex SQL queries, significantly improving data retrieval performance.
- Implemented observability using Datadog and AWS CloudWatch (metrics, logs, dashboards).
- Automated error log analysis with Python scripts, generating structured reports for debugging.

Arcesium **Jan 2023 - May 2023**
Site Reliability Engineer Intern *Hyderabad, IND*

- Integrated monitoring systems with Datadog for real-time metrics and visualization.
- Migrated workloads from legacy orchestration (Podmaster) to modern platforms (Veritas, Veda).
- Identified and reduced service latency by optimizing HTTP request flows.
- Resolved bugs, and maintained infrastructure using **AWS**, **Linux**, **Python**, and **GitLab CI/CD** pipelines.

Bharat Electronic Limited **May 2022 - Aug 2022**
Internship *Hyderabad, IND*

- Developed GNSS receiver algorithms (RAIM, TRAIM, multipath mitigation), improving accuracy by ~20%.
- Applied signal processing techniques for fault detection in satellite navigation systems.
- Contributed to defense R&D (CRL-BEL) with real-world GNSS experimentation and testing.

EDUCATION

University College of Engineering (Autonomous), OU **Graduation: May 2023**
B.E. in Computer Science and Engineering *Hyderabad, IND*

PROJECT

VERA — Enterprise Knowledge Brain | RAG & Hybrid Search | Github
Tech Stack: Python, RAG, FAISS, BM25, OpenAI Embeddings, Anthropic Claude, SQLite

- Developed an enterprise knowledge assistant using Retrieval-Augmented Generation (RAG) to retrieve contextual information from technical documentation, code artifacts, and engineering knowledge sources.
- Engineered a document ingestion and indexing pipeline with text preprocessing, intelligent chunking, metadata enrichment, embedding generation, and FAISS/BM25 indexing for semantic and lexical search
- Designed a hybrid retrieval architecture combining vector similarity search, Reciprocal Rank Fusion (RRF), and cross-encoder reranking, with query intent classification and source-grounded LLM responses.

ScribeAI — Generative AI Content Platform | LLM Integration | Github
Tech Stack: TypeScript, Node.js, Express.js, Groq LLM API, React, Stream Chat, Supabase

- Developed an LLM-powered content generation application supporting writing, summarization, rephrasing, and contextual responses through Groq API integration and a session-based chat interface..
- Implemented intent-aware prompt engineering with dynamic system prompts, configurable generation parameters, and conversation context management.
- Engineered event-driven AI request processing with webhook deduplication, rate limiting, session persistence, and safeguards against recursive AI responses.

CoreGPT — Transformer Language Model from Scratch | Github
Tech Stack: Python, Transformer Architecture, Natural Language Processing, Neural Networks

- Implemented a miniature GPT-style language model from scratch in pure Python, including tokenization, embeddings, multi-head causal self-attention, feed-forward networks, and transformer blocks.
- Developed model training and validation workflows using next-token prediction, cross-entropy loss, and gradient-based optimization.Automated CI/CD with GitHub Actions for testing, linting, and Docker builds.
- Implemented autoregressive text generation with temperature-based sampling to explore transformer inference and language modeling fundamentals.

GitInsight Engine — GitHub Analytics & Data Processing Platform | Github
Tech Stack: Node.js, REST APIs, GraphQL, Redis, Fastify, GitHub APIs

- Developed a modular GitHub analytics engine to collect and analyze repository activity, commit history, language distribution, code frequency, and contributor metrics.
- Implemented ETag-based conditional requests, TTL caching, and resilient API handling to reduce redundant requests and manage external API limitations.
- Designed configurable data aggregation and reporting workflows with structured JSON and SVG outputs for developer analytics and dashboards.